

Food AI Companion

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Milestone 3

1 Methodology

In Milestone 3, we extend the Food AI Companion into a more complete multi-mode recommendation system with personalization-aware retrieval, evaluation analysis, grounded conversational interaction, and expanded recommendation modes. This milestone specifically focuses on addressing the limitations identified in Milestone 2, including missing quantitative evaluation, incomplete secondary modes, limited grounding analysis, and the absence of filter-bubble experimentation.

The system now supports three integrated recommendation modes:

- **Eat Out:** NYC restaurant recommendations using retrieval-augmented generation and live restaurant metadata.
- **Cook at Home:** Pantry-aware recipe generation conditioned on user preferences and available ingredients.
- **Cocktails:** Flavor- and inventory-based cocktail recommendations grounded in CocktailDB data.

Additionally, we introduce a conversational Food Companion interface capable of routing user requests across modes while maintaining context from the active recommendation state.

1.1 Personalization and Retrieval Improvements

We extend the retrieval system from Milestone 2 by incorporating learned cuisine preference weights directly into both embedding-query construction and reranking. User accept/pass feedback updates cuisine scores over time, allowing retrieval outputs to adapt dynamically to evolving user preferences.

To improve robustness, we replace the original strict filtering logic with a three-stage fallback pipeline:

1. Strict cuisine and food filtering
2. Cuisine-only filtering
3. No filtering fallback

This prevents empty recommendation outputs while preserving relevance whenever possible.

We also expanded category-food mappings to reduce mismatches between cuisine labels and inferred food preferences across additional cuisines including Chinese, Mexican, Thai, Indian, and American categories.

1.2 Filter Bubble and Evaluation Analysis

A major focus of Milestone 3 is the implementation of quantitative personalization analysis.

We introduce a filter-bubble evaluation framework that measures:

- Recommendation overlap
- Novelty
- Diversity
- Entropy
- Profile alignment
- Weighted profile alignment
- Filter bubble index

The weighted alignment metric uses learned cuisine and food preference scores rather than binary matching, allowing personalization strength to evolve continuously with user feedback.

We also implement a deterministic evaluation harness that produces system-side metrics and standardized output summaries while avoiding fabricated human evaluation scores. Human evaluation templates are generated separately for optional manual review.

1.3 Cook at Home Mode

The Cook at Home mode is expanded into a complete recommendation pipeline that uses:

- Pantry ingredients
- Cuisine preferences
- Disliked foods
- Budget constraints
- Occasion context
- Time constraints

Because no structured recipe retrieval dataset is available, this mode is implemented as an LLM + taste-profile generation pipeline rather than full recipe RAG. Generated recipes include pantry usage transparency, ingredient lists, cooking steps, and remix/save/pass interactions.

1.4 Cocktail Grounding

The Cocktail mode uses CocktailDB grounding combined with LLM-based ranking and formatting. Recommendations incorporate:

- Flavor/vibe requests
- Bar inventory matching

- Ingredient overlap
- Saved taste-profile context

Each cocktail recommendation includes an explainability panel displaying the grounding source, inventory matches, request context, and known limitations.

1.5 Conversational Food Companion

We introduce an agentic conversational companion capable of:

- Inferring user intent across modes
- Switching between Eat Out, Cook, and Cocktail modes
- Asking clarifying questions
- Confirming actions before execution
- Maintaining active recommendation context
- Performing retrieval-grounded conversational recommendations

The conversational layer implements lightweight conversational RAG by retrieving relevant restaurant entries from the NYC dataset during interactions, allowing the Food Companion to ground conversational recommendations in retrieved evidence while dynamically routing users across Eat Out, Cook, and Cocktail modes.

2 Results

Milestone 3 substantially improves both personalization quality and system completeness compared to Milestone 2.

2.1 Eat Out Recommendations

The Eat Out pipeline produces grounded restaurant recommendations using personalized retrieval and reranking. Recommendations now include:

- Live restaurant metadata
- Cuisine-aware reranking
- Taste-profile alignment
- Review-aware explanations
- Opening-hour information
- Recommendation trace panels

For conversational queries such as “I want something cozy for tonight,” the system successfully routes the request into the Eat Out pipeline and retrieves contextually aligned restaurants with supporting explanations.

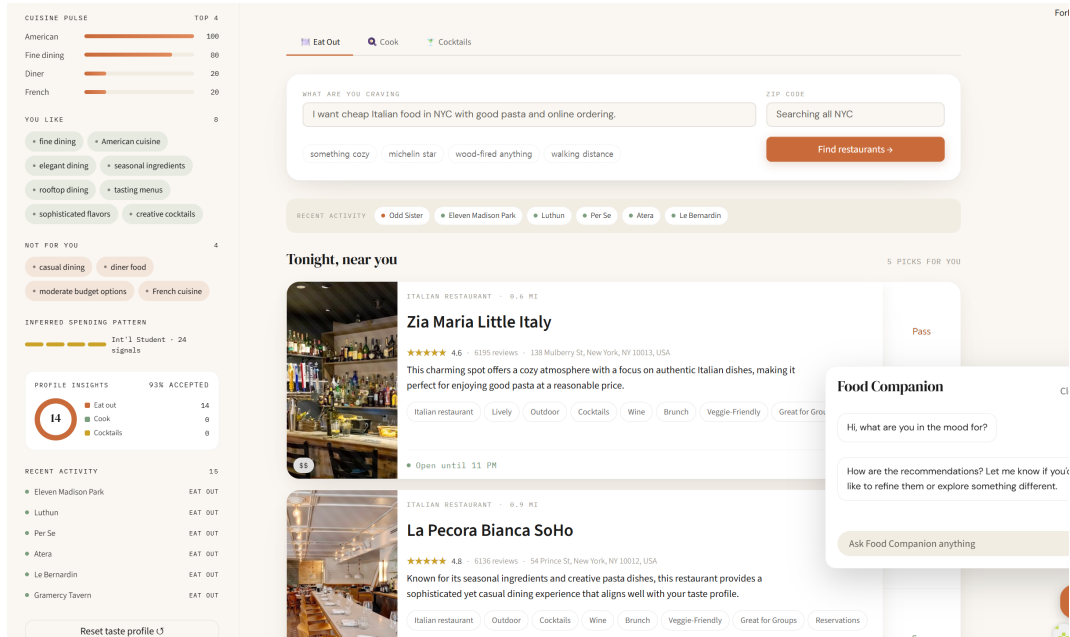


Figure 1: Grounded Eat Out recommendation results with conversational routing.

2.2 Cook at Home Results

The Cook mode successfully generates pantry-aware recipes conditioned on both available ingredients and user preference signals.

For example, given the pantry:

“eggs, rice, spinach, soy sauce, garlic”

the system generates a Korean-inspired fried rice recipe with transparent pantry usage, ingredient lists, and cooking instructions.

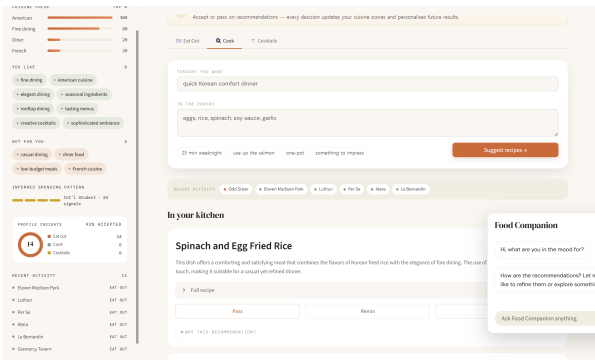


Figure 2: Cook at Home recommendation interface.

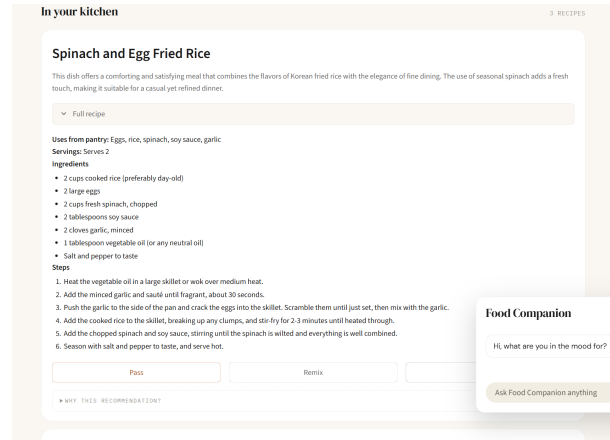


Figure 3: Expanded pantry-aware recipe generation output.

2.3 Cocktail Recommendations

The Cocktail mode produces grounded cocktail recommendations using CocktailDB evidence combined with inventory matching and vibe-based ranking.

Explainability panels provide transparency regarding:

- Grounding source
- Ingredient overlap
- Inventory matches
- Request context
- Recommendation limitations

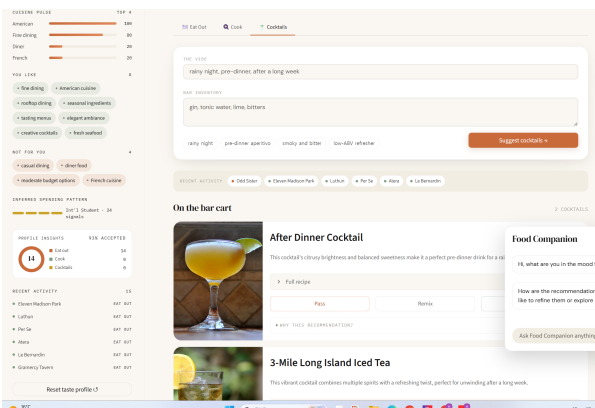


Figure 4: Cocktail recommendation interface with vibe-based retrieval.

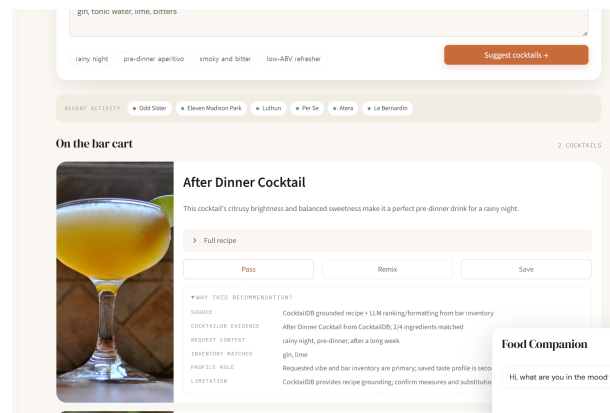


Figure 5: Cocktail explainability panel with grounded evidence.

2.4 Conversational RAG Interaction

The conversational Food Companion supports retrieval-grounded dialogue and cross-mode interaction. Users can issue vague conversational requests such as “I want something cozy for tonight,” and the system dynamically routes the request into the appropriate recommendation mode while retrieving relevant restaurant evidence.

The companion maintains context from the active recommendation state and supports conversational follow-up interactions for refinement and clarification.

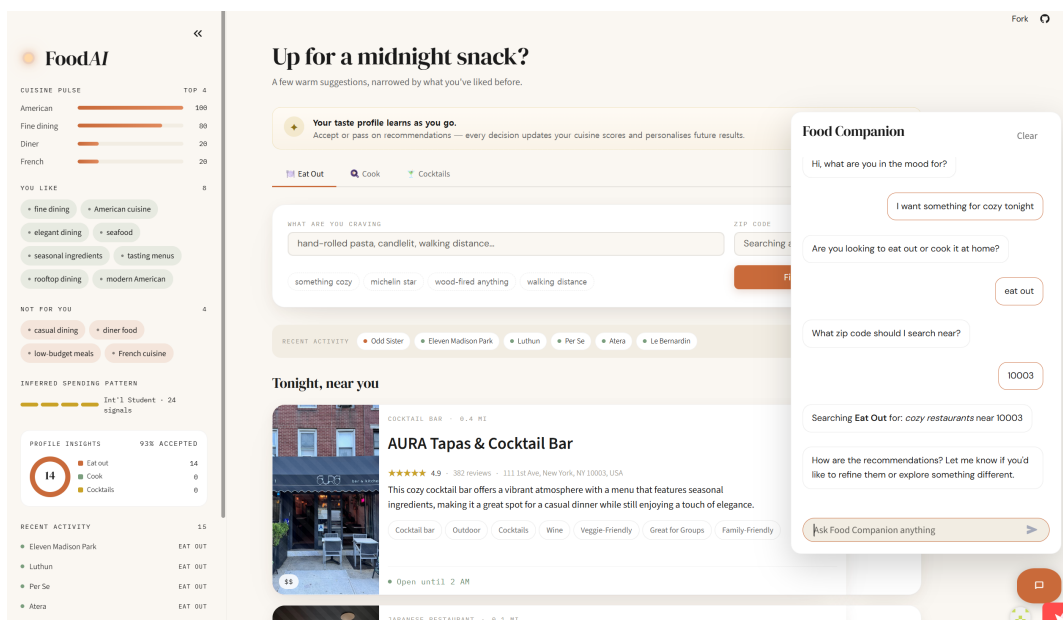


Figure 6: Conversational Food Companion interface with agentic routing and retrieval-grounded recommendations.

2.5 Quantitative Evaluation

Our evaluation framework introduces quantitative metrics for personalization and diversity analysis.

Results indicate that stronger personalization improves profile alignment while also increasing the possibility of recommendation concentration over time. Diversity-aware reranking partially mitigates this effect by introducing small bonuses for novel but profile-compatible results.

The filter-bubble analysis demonstrates that recommendation overlap increases after repeated accept-feedback sessions, supporting our original hypothesis that personalization strength and diversity may trade off against one another.

3 Analysis

The results demonstrate several important improvements over Milestone 2.

First, the system now supports significantly stronger grounding and transparency. Recommendation trace panels expose retrieval evidence, inventory matching, and grounding sources directly within the interface, improving explainability and reducing unsupported recommendations.

Second, personalization behavior is substantially improved through weighted cuisine scoring and feedback-aware reranking. Recommendations evolve over time as users interact with the system,

creating more adaptive outputs than the static profile logic from Milestone 2.

Third, the system now supports complete multi-mode interaction. Cook at Home and Cocktail modes are no longer placeholder interfaces and instead provide fully interactive recommendation pipelines with save/pass/remix behavior.

The conversational Food Companion also improves usability by allowing users to issue vague or incomplete requests that are clarified and routed automatically into the appropriate recommendation mode.

However, several limitations remain.

The Reddit/PRAW grounding artifact currently functions primarily as an offline analysis component rather than a fully integrated live retrieval source. Additionally, the Cook mode relies on generated recipes rather than retrieval-grounded recipe datasets due to limited structured cooking data.

Although the evaluation framework introduces quantitative metrics, the evaluation remains partly proxy-based and does not yet include large-scale human user studies.

Finally, personalization introduces an inherent trade-off between alignment and diversity. While stronger preference learning improves relevance, it may also reduce novelty and increase recommendation repetition over time.

4 Conclusions

Milestone 3 transforms the Food AI Companion from a partially complete recommendation prototype into a more integrated and evaluation-aware personalized recommendation system.

Key strengths of the final system include:

- Multi-mode recommendation support
- Retrieval-grounded personalization
- Explainable recommendation traces
- Conversational interaction
- Quantitative filter-bubble analysis
- Pantry- and inventory-aware recommendation generation

At the same time, several important limitations remain:

- Limited external live grounding sources
- Partial reliance on generated content
- Proxy-based evaluation methodology
- Potential long-term filter bubble effects

Future work could extend the system with:

- Live Reddit/forum retrieval integration
- Larger recipe retrieval datasets

- Long-term memory architectures
- Real-user evaluation studies
- Stronger diversity-aware recommendation optimization

5 Workflow and Contributions

Week 1 – Evaluation and Error Analysis

Owen Nie

- Define evaluation metrics for recommendation quality (relevance, alignment with user preferences)
- Design comparison framework across baseline, profile-based, and RAG systems

Hoerim Kim

- Analyze retrieval outputs and identify common failure cases
- Evaluate diversity and novelty of recommendations

Bilal Naseer

- Implement evaluation pipeline and organize output logs for analysis
- Standardize output formats for easier comparison

Week 2 – Retrieval and RAG Improvements

Hoerim Kim

- Improve retrieval query construction using richer taste profile signals
- Refine embedding and similarity scoring

Owen Nie

- Strengthen reranking and filtering logic to reduce mismatches
- Analyze trade-offs between precision and diversity

Bilal Naseer

- Improve prompt design to better utilize retrieved evidence
- Reduce hallucination by enforcing stricter grounding in RAG outputs

Week 3 – External Data Integration

Hoerim Kim

- Integrate additional data sources (e.g., Reddit or extended review data) into retrieval

Owen Nie

- Incorporate external signals into ranking and evaluation

Bilal Naseer

- Update RAG pipeline to incorporate external data into prompts

During Milestone 3, team contributions were divided as follows.

Week 4 – System Extension, Conversational RAG, and Final Improvements

Bilal Naseer

- Expanded Cook at Home mode into a full pantry-aware recommendation pipeline
- Implemented recipe remixing, save/pass feedback behavior, and explainability panels
- Added evaluation harness and filter-bubble analysis pipeline

Owen Nie

- Implemented the conversational Food Companion chat interface
- Added conversational RAG retrieval using the NYC restaurant dataset
- Enabled agentic routing across Eat Out, Cook, and Cocktail modes
- Redesigned result-card interfaces and interaction behavior across all modes

Hoerim Kim

- Improved personalization-aware retrieval using learned cuisine preference scores
- Extended diversity-aware reranking and fallback retrieval logic
- Validated personalization, diversity, and evaluation behavior across modes

All Members

- Conduct final system testing and integration
- Refine Streamlit UI and final demo interactions

Week 5 – Final Report and Presentation

All Members

- Prepare final report with complete analysis and results
- Create presentation slides and demo materials
- Conduct rehearsals and refine system demonstration